

Effect of microplatelet orientation in 3D printed microplatelet reinforced composites with bioinspired microstructures

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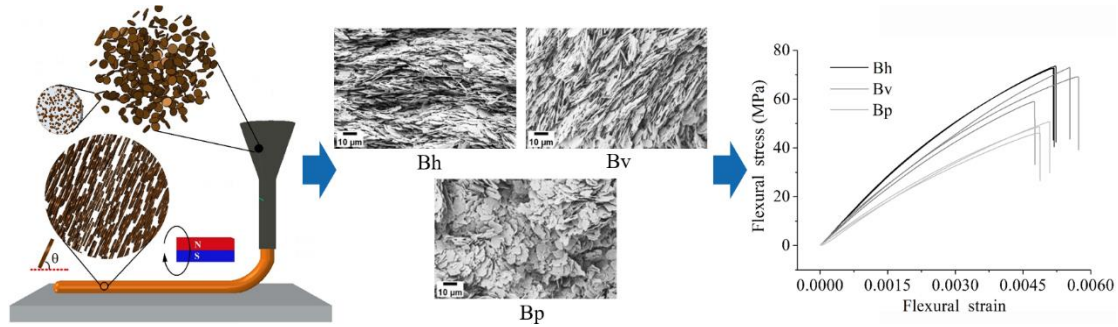
Abstract

Complex microstructures are the hallmark of natural ceramic biocomposites, but limited processing methods to reproduce them hinder the understanding of mineral orientation roles on the mechanical properties. This study investigates the influence of microplatelet orientation in composite materials, utilizing the magnetically assisted direct ink writing method (M-DIW) to create microstructured microplatelet-reinforced composites. Experimental and computational approaches are employed to explore the critical role of microplatelet orientation on the flexural properties of these materials. Horizontal microplatelets are found to significantly enhance the composite's flexural toughness by promoting overlap and increasing fracture energy during crack propagation. Vertical microplatelets contribute to increased flexural modulus and strength. Perpendicular microplatelets facilitate straight crack paths and smoother fracture surfaces. Moreover, complex microstructural designs were introduced by strategically combining microplatelet orientations to optimize mechanical properties. These findings emphasize the vital role of microplatelet orientation in composite materials, offering potential for tailored materials with superior performance.

Impact statement

The findings of this research carry significant implications in the fields of material science and engineering. By comprehensively examining the role of microplatelet orientation in composite materials, this study offers a novel perspective on how to optimize mechanical properties for various applications. The identification of distinct strengths and limitations associated with horizontal, vertical, and perpendicular microplatelet orientations enables the creation of tailored materials with enhanced mechanical performance. This customization potential holds considerable promise for industries that rely on composite materials, such as aerospace, automotive, and construction. Moreover, the introduction of hierarchical designs, presents innovative avenues for engineering materials with superior properties. These designs showcase the potential to achieve a delicate balance between flexural toughness, strength, and modulus, allowing for materials that can outperform traditional monolithic structures. Ultimately, this research empowers material scientists and engineers to make informed decisions regarding microplatelet orientation, enhancing the efficiency and versatility of composite materials across a wide range of industries. As a result, it brings us one step closer to a future where materials can be precisely tailored to meet the demands of specific applications, driving innovation and progress in diverse sectors.

Graphical abstract



Key words: 3D printing; reinforced composite; microstructure; simulation; mechanical properties

1| Introduction

Natural structural materials have evolved complex microstructural designs over billions of years of evolution, giving rise to a diverse range of intricate hierarchical microstructures that yield exceptional performance in terms of strength, resilience, toughness, and adaptability.^{1, 2} These natural microstructures serve as inspiring examples for the development of advanced materials, technologies, and architectural designs.³⁻⁵ Amongst the many natural microstructures, the microplatelet-based brick-and-mortar arrangement found in the nacreous layers of seashells is the most studied.⁶ However, synthetically reproduced nacre composites always show local misalignment of the microplatelets,⁷ and little is known about the effect of this misalignment on the final properties. Furthermore, there exists a number of more complex microstructural alignments, such as the concentric patterns found in trabecular bone,⁸ the layered structure of teeth,⁹ or the cross-lamellar structure in mollusk shells,¹⁰ but it is difficult to understand the role of the minerals orientations due to the lack of processing methods to reproduce them in artificial composites. Understanding the effect of microplatelet orientation on the macroscopic properties of reinforced composites would be interesting to then design microstructures specific for an intended application.

Bioinspired microstructural designs have been typically explored using 3D printing of hard and soft polymeric segments in various patterns.¹¹ It was found that twisted plywood and concentric hexagonal patterns exhibit the highest work of fracture, whereas lamellar structures provide only a small improvement on the toughness. Furthermore, by combining the various patterns, it was possible to tailor the properties of the macroscopic materials to produce materials that are strong and compliant, strong and tough, flexible and fast morphing response.¹² Focusing on a specific design, multimaterial 3D printing with hard and soft polymers revealed that toughness amplification can be realized using bone-inspired design.¹³ Although these works demonstrate the benefits of microstructuring polymers, these materials do not present the level of performance of ceramic-reinforced composites due to their lack of hard, mineral elements. In addition, ceramic-reinforced composites display fracture mechanisms including ceramic-matrix debonding, crack bridging, and others, that cannot be reproduced in purely polymeric systems. To enable similar type of study of the impact of microstructural designs on the mechanical properties of ceramic microplatelet-reinforced composites, an appropriate

82 fabrication method would be therefore needed. Furthermore, because reinforced composites
83 typically contain particles such as microplatelets, that are of micron-size, the microstructural
84 designs to explore would be scaled down: in the soft and hard polymer systems, the distance
85 between the soft and hard phases is typically a few hundred micrometers whereas in the ceramic
86 composite system, it would be a few micrometers. With such microplatelet-reinforced
87 composites, the effect of alternating layers with independent microplatelet orientation within
88 each layer has been studied where a hybrid nacre-like helicoidal structure with angle of 90°
89 between each layer exhibits a tradeoff between strength and toughness under compression.¹⁴
90 To explore more complex microstructural designs, curved layers have been produced so that
91 the microstructure not only varies along the z axis but also within the x-y plane.¹⁵ This
92 additional variation in the microstructure was found to further increase the strength and
93 toughness of the composites. To better understand the effect of microplatelet orientation on the
94 properties of the composites, other and more complex microstructural designs would also be
95 needed.

96 Various 3D printing strategies have been employed to fabricate bioinspired microplatelet
97 reinforced composites with complex microstructures by combining 3D printing with external
98 fields to control the microplatelets' orientation. For example, using the direct ink writing (DIW)
99 3D printing method, microplatelets dispersed in a liquid polymer ink can be extruded and
100 aligned along the flow direction due to shear forces at the extruding nozzle, albeit the
101 orientation cannot be tailored as desired.^{16, 17} Using digital light processing (DLP) in
102 conjunction with external magnetic or acoustophoretic fields, composites with local
103 microplatelets orientation can be realized, but only in diluted systems.^{18, 19} In contrast, by
104 employing ultra-low rotating magnetic fields along with a drop deposition method, aligned
105 microplatelet reinforced composites with voxelated microstructural control can be fabricated.²⁰
106 Because this method has low efficiency, another similar approach uses ultra-low rotating
107 magnetic fields with DIW and a porous substrate as the print bed.²¹ The inks are aqueous based
108 and optimized so that the microplatelets can align at any desired angle while the ink is drying
109 and the nozzle is moving forward. The method has demonstrated the fabrication of
110 microstructured materials leading to anisotropic properties after sintering but has not yet been
111 used to explore in depth the effect of microstructural design on the mechanical properties.
112 Indeed, the sintering step leads to large shrinkage which causes defects and deformations.
113 Therefore, more efficient approaches that allow for local changes in microplatelet orientations
114 and the creation of complex architectures hold significant importance for further advancements
115 in mechanical applications.

116 In this study, we use 3D printing to produce microstructured microplatelet-reinforced
117 composites and study the influence of the microplatelets' orientations on the macroscopic
118 mechanical properties of the composites using experiments and computational modeling. First,
119 we used the magnetically assisted direct ink writing method (M-DIW, which has been utilized
120 in a same context in previous studies by our group²¹) to print continuously a liquid slurry
121 containing a high concentration of magnetically functionalized ceramic microplatelets, using a
122 porous substrate as the print bed. This slurry is then used to produce microstructured composites
123 layer-by-layer. As we examine the microstructure of the printed materials, we observe
124 misorientations at the edges of each filament, which can be attributed to shear forces acting
125 during the printing process. To gain a deeper understanding of the impact of horizontal layers

on the flexural properties of the materials, we employ finite element (FE) simulations. The simulation results are then validated by experiments. In the final part of this work, we propose potential hierarchical designs with local orientations of microplatelets that can lead to composites achieving specific material properties, as predicted through our research and analysis. This approach opens up exciting possibilities for the design and fabrication of advanced materials with tailored microstructures and enhanced mechanical properties.

2| Materials and methods

2.1| Experimental

2.1.1| Slurry preparation and characterization

Alumina microplatelets (Ronaflair, Merck), with an average diameter of 8.5 μm and a thickness of 300 nm, underwent a magnetization process as previously described.¹⁴ In summary, a suspension of 0.01 vol% iron oxide nanoparticles (EMG-705, 10 nm in diameter, Ferrotec) and Al_2O_3 microplatelets was prepared in water at a neutral pH and stirred until complete electrostatic adsorption of the Fe_3O_4 nanoparticles onto the microplatelets occurred. The resulting magnetic microplatelets, referred to as m- Al_2O_3 , were subsequently separated through filtration and thoroughly washed with water and ethanol before being dried. These functionalized m- Al_2O_3 microplatelets were then utilized in the slurry. The slurry consisted of 20 vol% m- Al_2O_3 dispersed in water with a few drops of Dolapix CE64 (Zschimeer & Schwarz, Germany). After ultrasonication (Sonopuls HD 4100, Bandelin, Germany) for a total of five cycles, each lasting 2 minutes at 20% amplitude, 5 wt% poly (vinyl pyrrolidone) (PVP, MW = 360,000, Sigma-Aldrich, China) was introduced into the slurry and thoroughly mixed for several hours.

2.1.2| 3D magnetic printing (M-DIW)

The 3D magnetic printing setup consisted of a commercial 3D Printer (Ender-3 V2, Creality, China) and a custom-made magnetic system in which a commercial neodymium magnet (N52, SG magnetics) was affixed to a DC geared motor (MS 3010D, Maisheng, China)). The specific positioning of the magnet setup depended on the desired orientation for the microplatelets within the ink. A rotating magnetic field with field strength of 5-50 mT was generated by placing the magnet at 1 cm from the printing nozzle tip. For the printing process, the ink was loaded into 5 mL syringes (Terumo syringe, Terumo Corporation) equipped with 0.41 mm diameter tapered tip nozzles (Nordson EFD) and inserted into the 3D printer. The computer aided design (CAD) model was designed and sliced using PrusaSlicer 2.6.1 (MK3S MMU2S, Czech Republic). The printing speed was consistently set at 2 mm/s, and the thickness for each printed layer was configured at 0.2 mm. Material extrusion was achieved by controlling the printer's piston motion through flowrate adjustment. All the printing parameters can be found in Supplementary Information (SI) Table S1. A porous gypsum substrate was used to expedite draining of the liquid present in the slurry and drying of the extruded ink during the printing. The gypsum substrate was made by dispersing 8 g of gypsum powder (Ceramix standard; Germany) in 8 g of water, let it set in a flat plastic mould, then drying it in an oven at 48 $^{\circ}\text{C}$ overnight. The resulting gypsum exhibited a total porosity of 60% measured using

Archimedes, and a density of 1.2 g/mL.

2.1.3| Microplatelet orientations characterization

After the M-DIW, the samples on their gypsum substrates were dried for 48 hours at 48 °C in an oven (IKA oven model 125 control dry, Malaysia), then the samples were carefully removed from the substrates. This was achieved by gently inserting a spatula along one side of the sample to release it from the substrate. Furthermore, fracture surfaces of the cross-section of the samples were imaged with a scanning electron microscope (JSM-5600LV SEM, JEOL Asia Pte Ltd, Singapore) after being sputtered with a layer of gold for 120 seconds. Image analysis of these patterns was carried out using Image J (NIH, USA) based on the SEM images. Orientation characterization of the microplatelets was performed using the software tool OrientationJ.

2.1.4| Infiltration of the scaffolds

After M-DIW and drying, the samples, which are porous scaffolds made of microplatelets bound together with the PVP, were infiltrated with epoxy (EM00300, Spurr Low-Viscosity Embedding Kit). This epoxy was chosen arbitrarily to demonstrate the possibility to fabricate reinforced composites and to measure their mechanical properties. First, epoxy resin was mixed with its curing agent, then the mixture was stirred for 2 min and degassed in a vacuum (Binder VD 53, Fischer Scientific Pte Ltd, Singapore) for 5 min. The liquid solution was then cast on the scaffold and placed in again vacuum for infiltration (Binder VD 53, Fischer Scientific Pte Ltd, Singapore) until no more bubbles were observed. The samples were then left to cure at 48 °C for 72 h (IKA oven 125 control dry, Malaysia). Excess epoxy surrounding the composites was peeled off. After curing, the composites were polished to standard dimensions of 40 mm * 6 mm * 2 mm for 3-point bending tests.

2.1.5| 3-point bending tests

3-point bending was done on a universal testing system, Instron 3366, with a 500 N load cell and at a loading speed 0.1 mm/min, until fracture. 5 samples of each configuration at least were fabricated and tested to calculate the average. Videos were recorded during the 3-point bending process using the optical microscope (Dinoscope) placed at 90° to the thickness of the samples. The mechanical values measured and calculated from the 3-point bending test are represented in SI Figure S1 A. The load P (N), displacement δ (m) from the universal tester, span length L (m) between supports, sample thickness h (m) and with b (m) were used to calculate the flexural modulus E_b (MPa), flexural strength σ_p (MPa), and flexural toughness G (kJ/m³) using the following formulas:

$$E_b = \frac{L^3}{4bh^3} \cdot \frac{\Delta P}{\Delta \delta} \quad \text{eq. 1}$$

$$\sigma_p = \frac{3P_{max}L}{2bh^2} \quad \text{eq. 2}$$

$$G = \int \sigma d\epsilon \quad \text{eq. 3}$$

where σ is the flexural stress $\sigma = \frac{3PL}{2bh^2}$, ϵ is the flexural strain $\epsilon = \frac{6h\delta}{L^2}$ as depicted in SI

Figure S1 B.

2.2| Computational modeling

2.2.1| Description of model

To analyze the impact of different platelet alignments and their corresponding proportions on the flexural mechanical properties, the finite element (FE) analysis software Abaqus 2020 was used to simulate the 3-point bending testing on composite models as shown in Figure 1. The model consisted of microplatelets (square-shaped, $10\ \mu\text{m} \times 10\ \mu\text{m} \times 500\ \text{nm}$) aligned in a specific direction, surrounded by a matrix. The matrix was meshed with solid elements (C3D4), and the microplatelet-matrix interface was simulated using cohesive elements. The sample was positioned between three cylinders $20\ \mu\text{m}$ diameter to emulate the 3-point bending test. The top cylinder was designated as a rigid body, with a reference point (RP) and an initial displacement moving downward to load the sample. The 'penalty contact' formulation with a friction coefficient was applied to address general contact throughout the loading process. The total composite sample had representative geometric dimensions of $126.5\ \mu\text{m} \times 26\ \mu\text{m} \times 21.5\ \mu\text{m}$ (length $\times$ height $\times$ width in the x, y, and z axes directions). To replicate the different microstructures produced during the experiment, this model was constructed using various layers containing horizontal, vertical, and perpendicular microplatelets. In this study, we calculated the volume percentage of the horizontal microplatelets (denoted as h) as L_1/L_0 , where L_0 represents the total height of the sample, and L_1 is the height of the layer containing horizontal microplatelets. The percentages examined were 0, 20, 40, 60, 80, and 100, which corresponded to the experimental conditions. Finally, the model was solved using Abaqus/Explicit to simulate the entire loading process, providing insights into the flexural properties based on different microplatelet alignments and proportions.

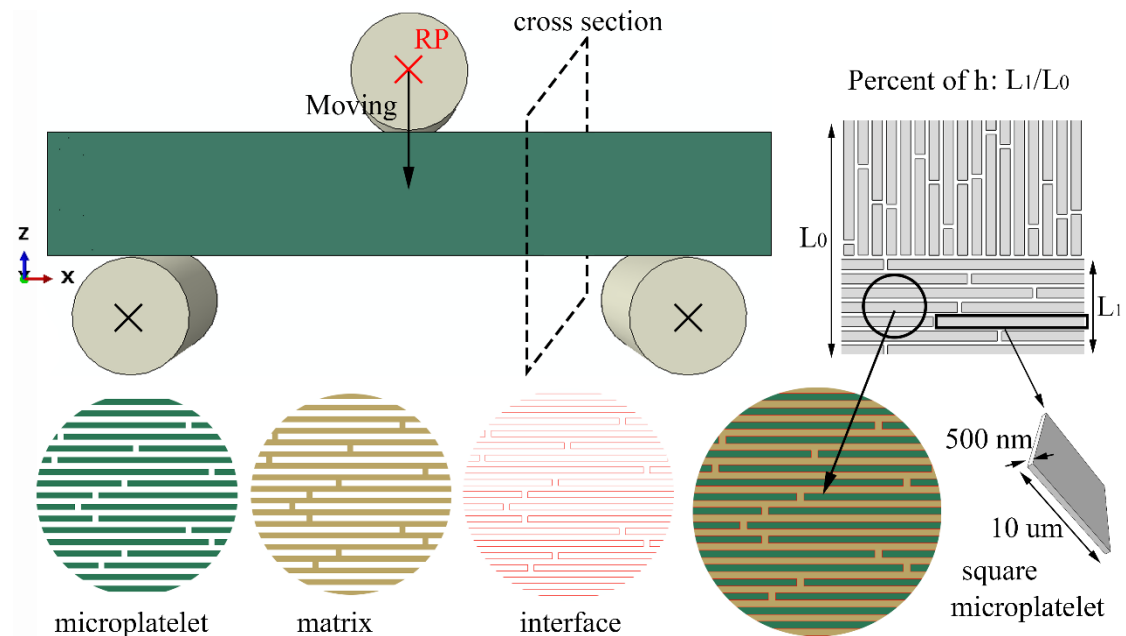


Figure 1: FE model of composites with aligned microplatelets. This model comprises three key components: the matrix, microplatelets, and interface. In the experimental context, the matrix corresponds to the epoxy material. The microplatelets are square-shaped, measuring $10\ \mu\text{m} \times 10\ \mu\text{m} \times 500\ \text{nm}$, simulating the characteristics of Al_2O_3 microplatelets. RP signifies the reference point

used for measurements, while L_0 denotes the total height of the sample, and L_I represents the height of the horizontal microplatelet layer within the composite structure. The ratio L_I/L_0 is used to indicate the volume percentage of the horizontal microplatelet layers.

2.2.2| Material modeling

In this study, only the elastic behavior was considered for the microplatelets since no fracture damage behavior of the microplatelets was observed in the experiment. The properties of the microplatelets were derived from the experiment, considering only their elastic properties. Continuum damage mechanics was used to describe the progressive softening in the matrix. As for the epoxy matrix, all property parameters were obtained and fitted from experimental stress-strain curves as shown in Figure S2. The discrete cohesive zone method was utilized to model the traction-separation behavior at the microplatelet-matrix interface, a constitutive method frequently employed. However, as reported in previous research on particle-reinforced epoxy composites,^{22, 23} it was observed that the interface properties were significantly higher than in our study. In particular, the normal strength and shear strength of interface were in the range of 50-120 MPa, even exceeding the strength of our epoxy matrix. Therefore, we made necessary adjustments to these interface properties to ensure their compatibility with the matrix strength. For reference, all the relevant material properties are reported in Table 1, providing a comprehensive overview of the properties that were considered during the analysis. Each failure mode is described in the following subsections.

Table 1 Mechanical properties of matrix epoxy, microplatelets and interface.

Material	Property
Microplatelet	Young's modulus $E_g = 377$ GPa, Poisson's ratio $\nu = 0.24$, Density = 3.95 g/cm^3
Matrix Epoxy	Young's modulus $E_m = 0.495$ GPa, Poisson's ratio $\nu = 0.31$, Density = 1.2 g/cm^3 Yield strength $\sigma_y^m = 36.23$ MPa, $\bar{\epsilon}_f^{pl} = 3\%$, Fracture energy $G_c^m = 0.001$ N/mm
Interface	Normal strength $t_n = 30$ MPa, Shear strength $t_s = 35$ MPa, Elastic stiffness $K = 100$ GPa/mm Fracture energy $G_I^C = 0.0002$ N/mm, $G_{II}^C = G_{III}^C = 0.0006$ N/mm Mixed-mode parameter: $\eta = 1.45$

2.2.3| Continuum damage model for epoxy matrix

The stress-strain curve obtained from uniaxial tension and compression tests, assuming the epoxy matrix exhibits isotropic and elastoplastic constitutive behavior is depicted in Figure S3. The linear elastic phase is characterized by the elastic modulus E_m . Upon reaching the yield strength σ_y^m , the classical plasticity constitutive model is employed to describe the yielding behavior of the matrix. The Mises yield surface is used to define isotropic yielding. It is defined by giving the value of the uniaxial yield stress as a function of uniaxial equivalent plastic strain. The plastic stage of the matrix is derived from an empirical curve fitting process utilizing experimental data,²⁴ as outlined in Table 1. The progressive damage and failure models were

used in conjunction with the plasticity. Damage initiation occurs when the equivalent plastic strain exceeds $\bar{\epsilon}_f^{pl}$. Subsequently, E_m decreases linearly until it reaches a saturation point at an equivalent plastic strain of $\bar{\epsilon}_f^m$, at which the matrix completely loses its load-bearing capacity. The fracture energy density corresponding to the shaded area g_c^m , multiplied by the characteristic length of the element l, is equal to the fracture energy G_c^m listed in Table 1.

2.2.4| Cohesive damage model for microplatelet-matrix interface

The interface between the microplatelet and the matrix has been modeled with cohesive elements, where the behaviors of these are defined in terms of a traction-separation law. The constitutive relation under pure tension (Mode I) or pure shear loading (Mode II) is shown in Figure S4. The initial response is assumed to be linear, with the slope representing the elastic stiffness K. Guidelines have been proposed for choosing this parameter:^{25, 26} generally it should be large enough to prevent interface interpenetration; overly large values would result in convergence difficulties; recommended values are in the range of 10–10⁵ GPa/mm.^{22, 27} In this work, a high value of K = 10² GPa/mm is applied in all three directions.²⁸ The shaded areas under the traction-separation curve correspond to G_I^C , G_{II}^C and G_{III}^C , which represent the fracture energies in the normal direction (Mode I) and the two local shear directions (Mode II and III), respectively.

Cohesive damage is assumed to initiate when:

$$\left\{\frac{t_n}{t_n^0}\right\}^2 + \left\{\frac{t_s}{t_s^0}\right\}^2 + \left\{\frac{t_t}{t_t^0}\right\}^2 = 1 \quad \text{eq. 4}$$

where t_n^0 , t_s^0 and t_t^0 denote the normal strength and the two shear strengths, respectively, and t_n , t_s and t_t are the nominal traction vectors. Due to the lack of experimental data, mixed-mode progressive damage is defined based on the interaction of fracture energies using the Benzeggagh-Kenane (BK) fracture criterion,^{29, 30} given by

$$G_I^C + (G_{II}^C - G_I^C) \cdot \left\{\frac{G_s}{G_T}\right\}^\eta = G^C \quad \text{eq. 5}$$

where $G_s = G_{II} + G_{III}$ and $G_T = G_I + G_{II} + G_{III}$, η is a mixed-mode parameter.³¹

3| Results and discussion

3.1| 3D printing and ink design strategy

To produce the microstructured microplatelet-reinforced composites and study the impact of local microplatelets' orientation on the mechanical properties, the recently established magnetically assisted direct ink writing (M-DIW) was used, albeit with a different ink than previously reported.²¹ Contrary to conventional DIW where viscous inks are used, the ink used in this paper exhibits liquid character and is an aqueous slurry containing a high concentration of ceramic microplatelets. The slurry is similar to the one used and optimized in a process called magnetically assisted slip casting (MASC),^{14, 15} where magnetically functionalized microplatelets are oriented with a ultra-low rotating magnetic field while the water from the slurry is getting removed through the pores of a porous mould, leading to high packing density while maintaining their orientation at any desired angle.³² After MASC, the dried samples can

be infiltrated with a polymer to yield a composite without the need for high temperature sintering, therefore ensuring the integrity of the set microstructure with no extra deformation. The M-DIW printing method was thus optimized to allow the preparation of microstructured composite beams for 3 points bending tests (Figure 2).

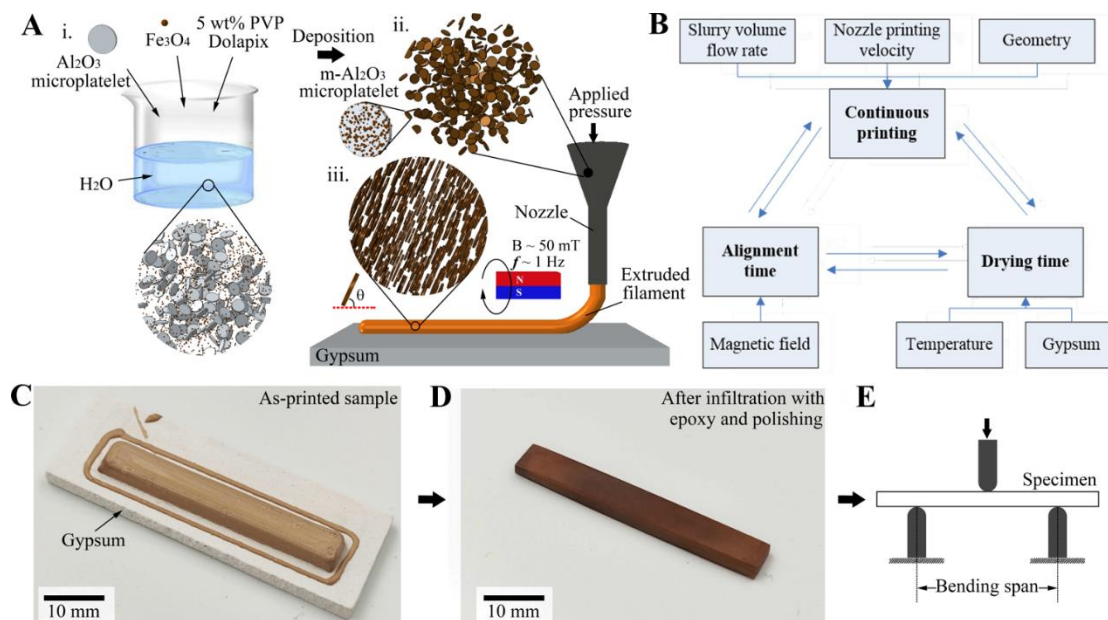


Figure 2: 3D printing of microstructured microplatelet-reinforced composites. (A) Schematics of the slurry preparation and 3D ink printing approach. (i) Illustration of the slurry preparation for the ink printing. PVP refers to poly (vinyl pyrrolidone). (ii) The magnetized alumina micro-platelets ($\text{m-Al}_2\text{O}_3$) after electrostatic adsorption of Fe_3O_4 nanoparticles on their surface as used in the slurry and will be oriented by the rotating magnetic field to create the microstructure. (iii) Aligned $\text{m-Al}_2\text{O}_3$ microplatelets retain their orientation after being magnetically oriented and deposited onto the porous gypsum substrate. (B) Optimization scheme for achieving a continuous 3D printing of microstructured porous materials, that involves balancing the alignment and drying times effectively. (C) Pictures of the as-printed samples on the gypsum substrate and (D) the samples after infiltration with epoxy and polishing to prepare the composite beam for testing. (E) Schematic diagram of the 3-point bending test.

To achieve M-DIW with our liquid slurry, it was crucial to find the right balance between the alignment time of the microplatelets under the rotating magnetic field, the ink drying time, and the continuous deposition process. Our ink slurry is similar to that previously developed for magnetically-assisted slip casting (MASC) which ensures quick microplatelet alignment within less than 6 s,³² and maintain of the microplatelets' orientations after drying. In brief, alumina microplatelets are made magnetically responsive through the decoration of their surface with iron oxide superparamagnetic nanoparticles (Figure 2A). These magnetically functionalized microplatelets align under low magnetic fields rotating at frequencies typically exceeding 1 Hz when the field strength is greater than 3 mT. The required frequency for alignment can be adjusted based on an established relation that considers the fluid's viscosity, the magnetic field strength, and the microplatelet's density, dimensions and magnetic susceptibility.³³ The slurry used for the ink contains 20% by volume of magnetic microplatelets

(m-Al₂O₃) dispersed with a surfactant and a small concentration of polymer (PVP) serving as a binder. The slurry exhibits shear-thinning properties and behaves like a liquid, which allows its extrusion and magnetic alignment of the microplatelets during printing. Simultaneously, the slurry solidifies, allowing multilayer printing after solvent drainage and drying through the porous substrate (Figure 2B).

After the preparation of the slurry, it is essential to select appropriate printing parameters for successful M-DIW. Achieving a balance between continuous ink printing, alignment time, and drying time is crucial. We achieved this balance by varying experimentally three critical parameters: the slurry volume extrusion rate, the nozzle printing speed, and the geometry of the printed samples. The extrusion rate, measured in $\mu\text{L/s}$, and the deposition rate, which quantifies the volume of slurry deposited per unit length of the drawn line ($\mu\text{L/mm}$), are of paramount importance. To achieve a complete part, the extrusion rate must fall within a narrow acceptable range. Under-extrusion leads to voids in the print, while over-extrusion results in rough top surfaces and potential deviations from the intended printing path (see SI Figure S5). Lower printing speeds give each layer more time to dry, leading to increased shrinkage and void formation. Inadequate adhesion of the slurry between deposited layers causes incomplete connections within the print (see SI Figure S6). The concentric path was found to be better, as it optimized the appearance of the outer surface by printing the outer vertical surface first, allowing it to dry first. This printing approach minimizes the accumulation of the wet slurry, as more deposited lines can dry before the next one is drawn. Additionally, it results in shorter travel moves of the nozzle (highlighted in blue in SI Figure S7), reducing overall travel time and minimizing anomalies at the ends of the printed object. Unlike other 3D printing techniques, the size of the printed part significantly influences the print quality in this method (see SI Figure S8). The size affects the layer drying time and, subsequently, influences the formation of voids due to increased shrinkage. In turn, the impact of layer height, particularly when adjusting the extrusion rate, is minimal in the range of 0.1-0.3 mm when the flow rate is appropriately adjusted (see SI Figure S9). However, using a larger layer height reduces the number of layers required to achieve the desired part height, resulting in faster printing. Furthermore, the magnetic field critically affects the microstructure inside the printed layers, which is what we aim to study here (see SI Figure S10). Without a magnet, microplatelet orientation is random. Exposure to a 1-20 mT or 5-50 mT rotating magnetic field predominantly causes vertical alignment in the cross-section. However, in the presence of a stationary magnetic field at 5-50 mT, most of the cross-section does not exhibit clear vertical alignment. Higher bed temperature leads to faster drying speeds which in turn accelerates the shrinkage of the slurry, leading to voids in prints if the extrusion rate is not properly adjusted to compensate for this rapid shrinkage (see SI Figure S11). To conclude, all the parameters employed for M-DIW printing in this study have been summarized in Table S1, with a representative final printed product depicted in Figure 2C.

After printing, the samples are weak and brittle since they are porous scaffolds constructed with microplatelets in various orientations and only bound together by a small amount of PVP. To make them stronger, the samples were therefore infiltrated with epoxy to create the final composites, as illustrated in Figure 2D. In the study, four epoxy candidates had been employed to assess the effects of infiltration, as depicted in Figure S12. Clearly, epoxy EM0300 stands out as the most effective, achieving full infiltration throughout the entire sample. In contrast,

the infiltration quality for the other epoxies is suboptimal; for instance, epoxy 932 and 941 exhibit incomplete infiltration, particularly at the center of the sample where some areas lack epoxy penetration. Consequently, epoxy EM0300 is the epoxy that was chosen in this work, resulting in an exceptionally low porosity and limited impact on the final mechanical properties. Furthermore, SEM images of samples with various configurations after infiltration with epoxy EM0300 are presented in Figure S13, revealing an absence of pores inside the samples. This indicates successful infiltration without resulting in any residual porosity. Additionally, the density of all samples is measured, as illustrated in Figure S14. The samples exhibit comparable density values, suggesting that the epoxy infiltration is nearly uniform across the different microplatelet configurations. After curing of the epoxy, the composites were polished to meet the standard dimensions of 40 mm * 6 mm * 2 mm for the 3-point bending tests as shown in Figure 2E.

In conclusion, achieving successful M-DIW printing demands balancing of factors like slurry properties, printing parameters, and magnetic field. Precise control is critical to ensure successful and high-quality, magnetically microstructured 3D prints. Then, we controlled the orientation of the magnetic field, which allowed us to obtain various microstructure composites, each exhibiting distinct mechanical properties. In the next section, we analyze the microstructures of the samples and their corresponding flexural properties.

3.2| *Microstructure analysis and flexural properties*

After 3D printing, the microstructures obtained under three different magnetic field orientations: Bh (horizontal), Bv (vertical), and Bp (perpendicular) were observed using electron microscopy and their mechanical properties measured using 3-points bending (Figure 3). The magnet in Bh configuration is placed so that microplatelets theoretically orient horizontally, parallel to the substrate. The magnet in Bv configuration is placed so that, theoretically, the microplatelets orient vertically, with their diameter oriented along the printing direction, and in Bp configuration, also vertically but with the microplatelets facing the printing direction. Bp and Bv are therefore perpendicular. Special attention was paid to the presence of horizontally aligned microplatelets after applying the magnetic fields in these configurations, because it is known that reinforced composites with horizontal microplatelets, also referred to as nacre-like composites, exhibit the highest strength and toughness as compared to composites containing microplatelets oriented at other angles.³⁴

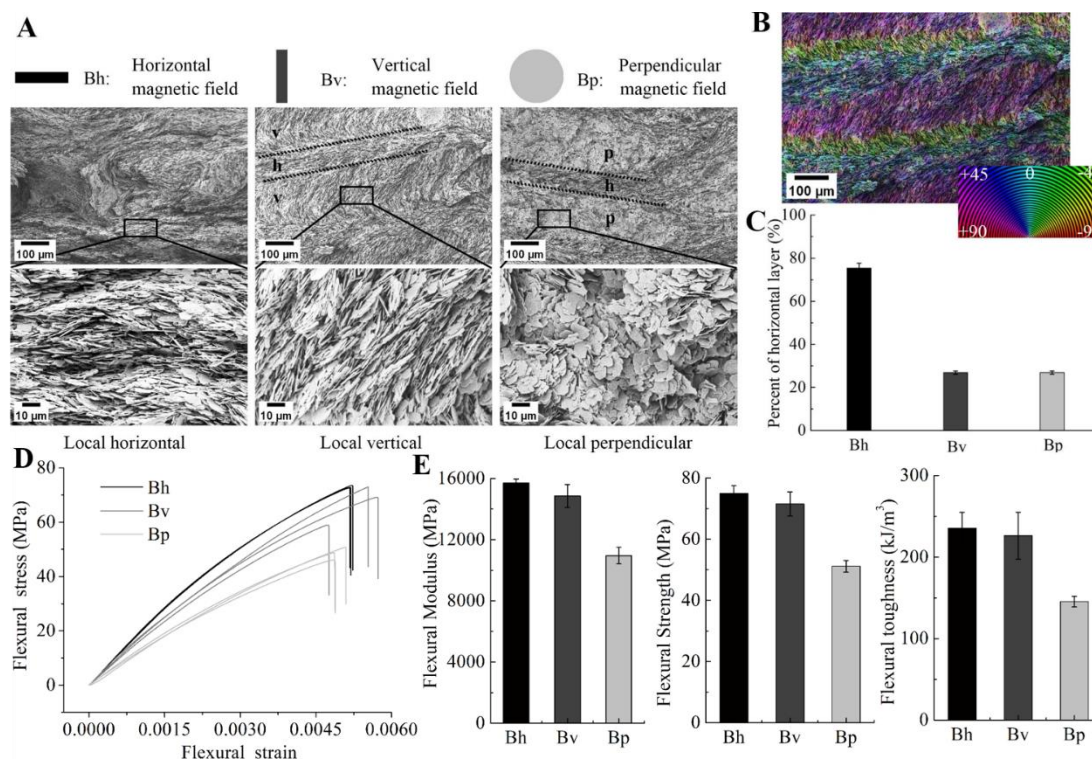


Figure 3: Microstructure and mechanical properties of the 3D printed composites. (A) Cross-sectional electron micrographs of samples under three different magnetic configurations: i. Bh (horizontal), ii. Bv (vertical), and iii. Bp (perpendicular), respectively. The cross-sections were obtained perpendicularly to the length of the sample beams shown in Figure 2C. (B) An image analysis method was used for characterizing the distribution of orientation as a function of angles degree. The insert picture represents varied orientations using a circular color map. (C) Obtained the percentage of horizontally aligned microplatelets in the cross-section of samples prepared under Bh, Bv and Bp configuration. (D) Typical flexural stress-strain curves obtained for the epoxy-infiltrated composites prepared with the 3 magnetic configurations, and (E) their flexural modulus, flexural strength and flexural toughness.

In general, the 3D printed samples prepared with each of the 3 different magnetic configurations did not display a monolithic microstructure with only one orientation, unlike previous work on microstructured layer composites (Figure 3A).^{14, 15} Instead, the samples exhibited diverse stacking layer orientations, particularly in the case of Bv and Bp. For example, under the vertical magnetic field (Bv), the microplatelets are stacked in a v-h-v-h-v pattern along the height direction, where v and h are layers with vertically and horizontally aligned microplatelets, respectively. A similar microstructural distribution is observed in the Bp samples, with layers arranged as p-h-p-h-p, where p and h are layers with vertically and horizontally aligned microplatelets, respectively, v and p differing according to Figure 3A. This local variation of the microplatelet orientations occurs due to shear forces acting at the edges of each filament during the printing process. Indeed, shear forces develop at the nozzle which align the microplatelets along the shear field in concentric, circular patterns.³⁵ To mitigate this influence, several approaches can be considered: reducing the viscosity of the slurry, decreasing the extrusion rate, employing a larger diameter nozzle, lowering the bed temperature, or substituting the gypsum substrate with a glass substrate. Each of these measures aims to

diminish the impact of shear forces. However, it is essential to note that implementing these measures may introduce challenges in achieving the solidification required for bulk printed samples. For instance, reducing viscosity can minimize shear forces during printing, but excessively low viscosity may hinder slurry solidification, impeding the seamless progress of continuous printing. Therefore, careful consideration is necessary when adopting these measures to ensure the successful implementation of magnetically assisted direct ink writing. When multiple lines are printed on top of each other, gravity compresses the circular shapes of the individual lines, leading to a rather square cross-section shape, and with microplatelets following the shape of the cross-section adding extra horizontal and vertical orientations in the final, dried sample.

Because microplatelets' misorientation is initially a microstructural defect, it raises an important question: what is the proportion of horizontal layers or vertical perpendicular layers in these samples? To answer this, we conducted a quantitative analysis to characterize the distribution of microplatelet orientations in terms of microplatelets angles (Figure 3B, C). Here, microplatelet angles in the range of -15° to 15° were defined as horizontal layers. The analysis revealed that samples produced in Bh configuration consisted of 75.3% horizontal layers, while in Bv and Bp configurations, they contained 26.9% horizontal layers each. These data suggest that the M-DIW 3D printing method allows for the adjustment and control of the microstructure, specifically the proportion of layers with horizontal microplatelet, ranging from 26.9% to 75.3%. Such control offers the potential to design and tailor the mechanical properties of the printed materials accordingly. Also, since 3D printing is a fabrication method that could be tentatively used to produce reinforced composites with tailored microstructure for a given application, better understanding what is the effect of these local misorientations is necessary to purposely control their mechanical properties.

The analysis of the microstructures in our 3D printed composites has revealed different stacking methods, prompting an inquiry into the influence of these stacking methods and the proportion of horizontal microplatelet layers on the mechanical properties. Consequently, 3-point bending tests were conducted to investigate the flexural properties of these 3D printed composites. The typical flexural stress-strain curves of the 3D printed composites exhibit an initial phase of linear elastic increase, followed by a nonlinear increase, and concluding with a sudden drop in the stress as the strain increases (Figure 3D). Subsequently, the flexural modulus, strength, and toughness were extracted from the stress-strain curves (Figure 3E). **In summary, composites prepared in the Bh configuration demonstrate higher flexural modulus, strength, and toughness compared to those in the Bp configuration that have the lowest values ($Bh > Bp$). This implies that a higher percentage of horizontal microplatelets yields superior flexural properties. Additionally, samples with vertical microplatelets (Bv) exhibit better properties than those with perpendicular microplatelets (Bp), even when they share the same percentage of horizontal layers. However, the distinction between Bh and Bv is not highly significant when standard deviations are taken into consideration. Consequently, to thoroughly explore the impact of platelet architecture on the observed differences, the FEM simulation is employed in the following section.**

In summary, the microstructure variations in 3D-printed composites under different magnetic field orientations were explored. Increasing the proportion of horizontal microplatelets significantly improves the flexural properties. Composites with vertical

microplatelets (Bv) outperform those with perpendicular microplatelets (Bp), showcasing the potential for tailoring the materials properties through 3D printing.

3.3| Finite element simulation on the flexural properties of the microstructure composites

The experimental results indicated that a higher proportion of horizontal microplatelets had the most significant positive impact on the flexural properties. However, the quantitative relationship between the proportion of horizontal microplatelets and the flexural properties remains unclear and understanding the roles of vertical and perpendicular microplatelets is still lacking. To address these questions, finite element simulation was employed to provide a deeper analysis (Figure 4).

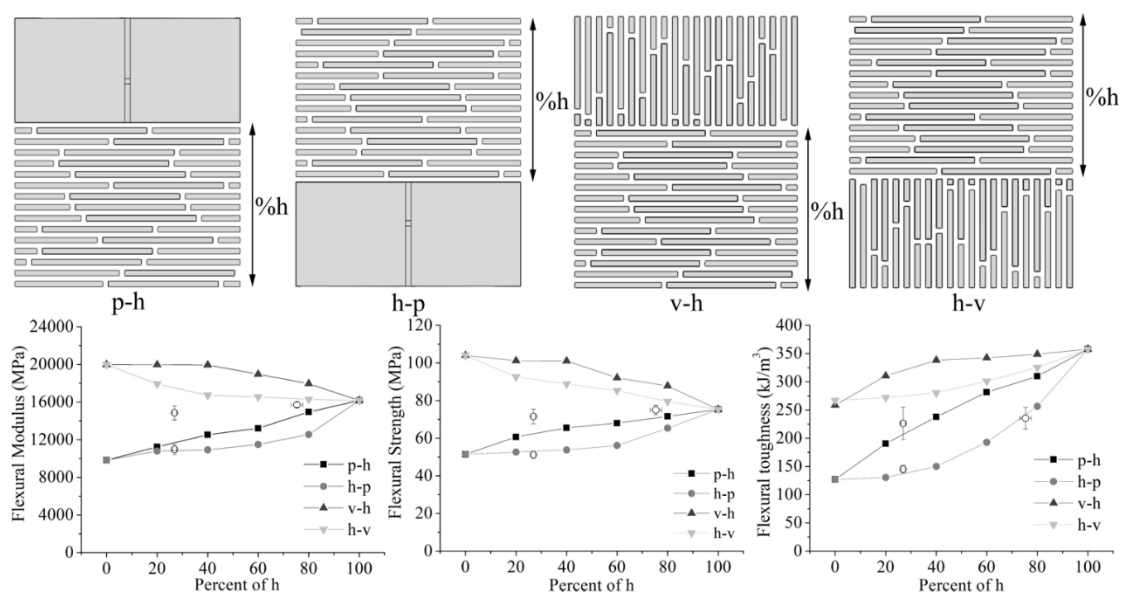


Figure 4: Finite element modelling: effect of the percentage of horizontally aligned microplatelets on the flexural properties. (A) Schematics showing the microstructure in the cross-section of the simulated p-h, h-p, v-h and h-v samples. The p-h, h-p samples refer to the combination of layers with vertically (p) and horizontally aligned microplatelets and with the h layer at the bottom and the top, respectively. The v-h, h-v samples refer to the combination of layers with vertically (v) and horizontally aligned microplatelets and with the h layer at the bottom and the top, respectively. **(B)** Flexural modulus, flexural strength and flexural toughness of the simulated p-h, h-p, v-h and h-v samples, respectively. The corresponding values from the experimental samples are represented with the empty circles and labelled Bv, Bp and Bh.

In the FEM simulations, samples containing layers with horizontal (h), vertical (v), and perpendicular (p) microplatelets were generated (Figure 4A). The effect of the percentage and location (top and bottom) of layers with horizontal microplatelets on the flexural properties was investigated. For the samples containing h and p microplatelets, a higher proportion of horizontal microplatelets led to improved flexural properties, including flexural modulus, strength, and toughness, in comparison to samples with perpendicular microplatelets. This effect is attributed to the overlap stacking of the horizontal microplatelets, resulting in longer jagged cracks propagating within the horizontal layer and creating rough fracture surfaces (see SI Figure S15A for the FE fracture model). As a result, more fracture energy is required during

the crack propagation in layers with horizontally aligned microplatelets as compared to the straight crack path and smooth fracture surface in layers with perpendicular microplatelets. In the case of a juxtaposition of layers containing v and h microplatelets, a higher percentage of vertically aligned microplatelets inversely resulted in higher flexural modulus and strength. However, a higher percentage of horizontal microplatelets in the samples still provided superior flexural toughness (Figure 4B). Furthermore, the presence of horizontal microplatelets at the bottom layer of the composite consistently improved the flexural properties. This can be attributed to the stress distribution during bending deformation (see SI Figure S16). To support this analysis, we constitute a theoretical analysis as following eq.6-8, aimed at illustrating the influence of height in the z-direction on the final flexural mechanical properties. Indeed, for any line $\overline{MN}$ crossing the width of the composite, the strain ε and stress σ can be obtained as:

$$\varepsilon = \frac{\widehat{PQ} - \overline{MN}}{\overline{MN}} = \frac{(\rho - z - h)d_\theta - \rho d_\theta}{\rho d_\theta} = -\frac{z - h}{\rho} \quad \text{eq. 6}$$

$$\sigma = E \cdot \varepsilon = E \left(-\frac{z - h}{\rho} \right) \quad \text{eq. 7}$$

where z is height of any line $\overline{MN}$, E is the young's modulus, ρ is the radius of curvature at the middle line of the cross section. h is the height at middle line of the sample. Here, it is assumed that there is no lengthening or shortening in the middle line. The bending moment M for any line $\overline{MN}$ can be obtained using:

$$M = F \cdot L = - \int \sigma \cdot d_A \cdot (z - h) = \frac{E}{\rho} \int (z - h)^2 d_A \quad \text{eq. 8}$$

As indicated by eq. 8, the bending behavior of any sub-volume within the cross section is directly related to its corresponding position in the height direction, but it remains unrelated to the other two directions. This fundamental principle explains why the positioning of a layer with horizontal microplatelets at the top or bottom of the composite leads to significant differences in mechanical response. In summary, vertical microplatelets tend to yield composites with better flexural modulus and strength, while horizontal microplatelets are associated with samples having the highest flexural toughness. This highlights the importance of microplatelet orientation. Furthermore, the positioning of the oriented layers also determines the specific mechanical properties of the microplatelet-reinforced materials.

To assess the reliability of the FE simulation results, the percentage of horizontal layers in experimental 3D printed samples (prepared using the three configurations, Bh, Bv, and Bp) was measured, and their data points were added to the simulated plots for comparison (Figure 4B). In general, the flexural modulus, strength, and toughness values obtained for the samples prepared using the Bh and Bp configurations match well with the values of the corresponding composites simulated in FEM, indicating a strong correspondence between the experimental and simulated data. This also suggests a high level of reliability of the FEM modeling approach used in this study. However, the data for the samples prepared in Bv configuration, containing vertical microplatelets, consistently falls below the corresponding simulated data for all flexural parameters. This discrepancy can be attributed to the ideal and perfect vertical microplatelet alignment assumed in the FEM model (Figure 4A), in contrast to the inevitable small misalignments of the vertical microplatelets observed in the experiments (Figure 3B). The distribution of microplatelet orientation as a function of degrees in Bv is illustrated in SI Figure

S17. Notably, the predominant alignment of microplatelets was found to be at approximately 50 degrees. In contrast to the ideal vertical alignment of microplatelets of 90 degrees, the misalignments represent deviations from the vertical orientation. Such microplatelet misalignments can result in a reduction in flexural properties, as demonstrated by the decrease in properties with the microplatelet tilt in monolithic composites (see SI plotted curve in Figure S18). Furthermore, perpendicularly aligned microplatelets exhibit a significant drop in properties compared to the other two configurations. This can be attributed to the higher volume fraction of the interface in perpendicular microplatelet composites. The weaker mechanical properties at the interface between the matrix and microplatelets contribute to the decline in flexural mechanical properties. Using the flexural modulus as an example as shown in Figure S19, the composite flexural modulus ($E_{composite}$) can be expressed generally as follows:

$$E_{composite} = E_m \cdot V_m + E_p \cdot V_p + E_f \cdot V_f \quad \text{eq. 9}$$

Here, E_m , E_p , E_f represent the flexural modulus of the matrix, microplatelets, and interface, respectively. Similarly, V_m , V_p , V_f denote the volume ratio of the matrix, microplatelets, and interface, with the constraint that $V_m + V_p + V_f = 1$.

For the interface of matrix and microplatelet, the volume fraction of the interface (V_f) for perpendicular microplatelets is considerably larger than that in horizontal or vertical microplatelets configurations as shown in Figure 3. Additionally, the modulus values follow the relationship $E_f < E_m \ll E_p$. Consequently, the composite flexural modulus ($E_{composite}$) for perpendicular microplatelets is lower than the moduli in horizontal or vertical microplatelets. Similar trends are expected for other flexural properties.

In summary, horizontal microplatelets in the composites improve the flexural modulus, strength, and toughness, while vertical microplatelets enhance modulus and strength. Positioning horizontal microplatelets at the bottom of the composites improved their overall flexural properties. Microplatelet orientation and positioning significantly affect mechanical properties, highlighting the importance of controlled microstructural design. **These simulated results however seem to confirm the trends seen in the experimental results (Figure 3E). Indeed, in the 3D printed samples, there is a stacking sequence of v-h for Bh and Bv, and h-p for Bp. As a rough approximate, the mechanical properties of Bh would be the average between the mechanical properties obtained for v-h and h-v configuration taken at 75.3% of h; for Bv, it will be similar but taken at 26.9 % of h; and for Bp it would be the average between p-h and h-p configurations taken at 26.9 % of h. In this simplified case, the flexural modulus of Bh, Bv, and Bp is respectively 17, 19 and 11.5 GPa, their flexural strength, 85, 95, and 56 MPa, and their flexural toughness 332.5, 300, and 170 kJ/m³. Therefore, there is also a very small difference in mechanical properties between Bh and Bv, while Bp has lower properties. It is therefore interesting to note that the volume fraction of horizontal layers may not have a dramatic effect at least within the range 25 to 75 %.**

3.4| Potential hierarchical design coupling with microplatelets local orientations

After analyzing the effects of composites containing layers with horizontal, vertical, and perpendicular orientations of microplatelets on the flexural properties, potential hierarchical designs can be obtained by incorporating this microplatelet orientation information (Figure 6).

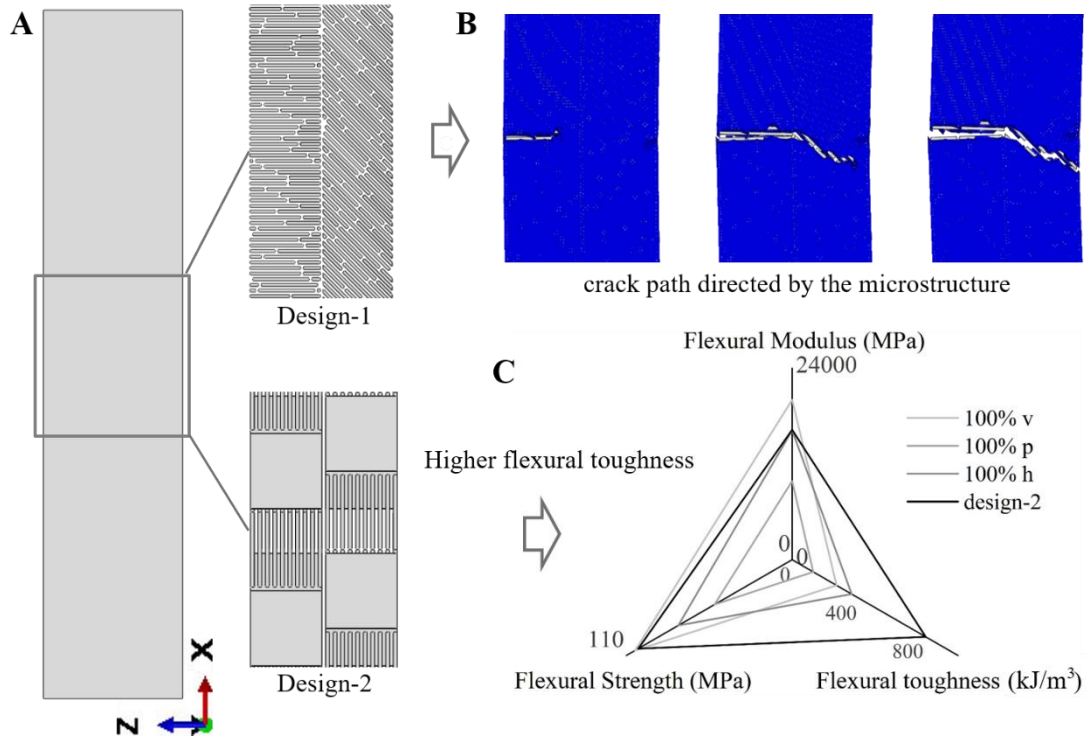


Figure 5: Potential hierarchical design with microplatelets local orientations. (A) Design-1 is composed of the perpendicular layer at bottom and its corresponding layer shifting round y-direction at top. Design-2 is the combination of the vertical and horizontal microplatelets overlapping in x- and z- directions. **(B)** The crack path propagation of design-1 along the alignment of microplatelets. **(C)** Comparison of design-2 with monolithic horizontal, vertical and perpendicular layers in flexural modulus, strength and toughness.

As discussed above, horizontal microplatelets offer the advantage of enhancing flexural toughness due to the microplatelets' overlap, requiring more fracture energy during crack propagation. Vertical microplatelets contribute to higher flexural modulus and strength but exhibit an average level of flexural toughness. Perpendicular microplatelets may not significantly improve the mechanical properties but display the advantage of promoting a straight crack path and a smooth fracture surface. In summary, each type of microplatelets has its own strengths and limitations. Therefore, through thoughtful design and application, it is possible to create hierarchical composites that not only maintain or exceed the original strengths of the microplatelets but also mitigate their weaknesses. This serves as the overarching design principle for hierarchical composite materials as shown in Figure 5.

Regarding the potential to achieve the desired fracture behavior or crack propagation, the use of perpendicular microplatelets offers the advantage of promoting a straight crack path. By employing only perpendicular microplatelets in Design-1, which consists of a perpendicular microplatelets layer at the bottom and a corresponding layer shifted around the y-direction at the top as depicted in Figure 5A, we successfully observed crack propagation along the platelet alignment as shown in Figure 5B. On the other hand, we are aware that horizontal layers contribute to high flexural toughness, while vertical microplatelets enhance modulus and strength. To explore the possibility of combining the benefits of both types of microplatelets to achieve higher modulus, strength, or greater flexural toughness, Design-2 was developed. In

this design, vertical and horizontal microplatelets are combined and overlapped in the x- and z-directions as illustrated in Figure 5A. The big grey squares extending along the z-direction correspond to the vertical microplatelets, whereas the smaller strip-like microplatelets aligned along the x-direction are designated as horizontal microplatelets. Design-2 has shown remarkable improvements in flexural toughness while still maintaining high flexural strength and modulus, surpassing the properties of monolithic horizontal, vertical, and perpendicular layers, as demonstrated in Figure 5C. These hierarchical designs leverage the unique characteristics of each platelet orientation, opening up new possibilities for advanced materials with tailored mechanical properties and superior performance.

4| Conclusion

In conclusion, we used the magnetically assisted direct ink writing method (M-DIW) to produce microstructured microplatelet-reinforced composites and explored the critical role of microplatelet orientation on the flexural mechanical properties by using experiments and computational modeling. We demonstrated that the presence of horizontal microplatelets in the composites can enhance the flexural toughness by promoting overlap and increasing the fracture energy during crack propagation. Vertical microplatelets, in turn, can boost the flexural modulus and strength of the composites, albeit with moderate toughness improvement. Finally, the presence of perpendicular microplatelets favors straight crack paths leading to smoother fracture surfaces. By strategically combining these microplatelet orientations in complex microstructural designs, it is possible to tailor and optimize the mechanical properties of microplatelet-reinforced composites. This research highlights the importance of microplatelet orientation in composite materials and offers exciting prospects for tailored materials with superior performance. Such insights drive advancements in material science and engineering, opening up new possibilities for the creation of innovative materials designed to meet specific industry and application requirements.

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Declaration of conflicts

The authors declare no conflict of interest.

Data availability

All data are available in the manuscript and in the attached supplementary file. Additional information can be requested from the authors.

Author contributions

W.P. - Writing - original draft, Visualization, Investigation, Data curation, Conceptualization.

X.Y.C. - Writing - review and editing, Data curation, Investigation.

S.H.L.- Writing - review and editing, Investigation.

H.L.F.- Writing- review and editing, Conceptualization, Supervision, Funding acquisition.

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